

Danny Kim

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Education

Occidental College

August 2022 - May 2026

Bachelor of Arts, Computer Science, Math minor GPA: 3.54

- Relevant coursework: Natural Language Processing, Deep Learning, Algorithms Analysis, Computer Architecture, Databases, Cryptography, Statistical Data Analysis, Numerical Analysis, Multivariable Calculus, Linear Algebra, Discrete Mathematics
- Extracurricular activities: COMP 101 Justice and Equity in Technology TA, COMP 131 Fundamentals of Computer Science TA
- Leadership activities: Computer Science Club (Head of Events/Treasurer), Korean American Student Association (President), Occidental Consulting Group (Analyst)

Skills

Languages: Python, Java, C++, SQL, JavaScript

Frameworks: PyTorch, Hugging Face Transformers, Scikit-learn, LightGBM, React, Next.js, Node.js, SHAP

Tools: Git, GitHub, MySQL, SQLite, Pandas, Anaconda, VS Code, PyCharm, Excel, PowerPoint

Projects

Network Intrusion GNN

Implemented a graph based anomaly detection model using PyTorch

- Modeled network traffic as a graph and implemented a GNN (Graph Neural Network) classifier in PyTorch to detect anomalous connections indicative of network intrusion
- Implemented an inductive learning pipeline to support generalization to unseen network nodes
- Designed modular anomaly detection pipelines, separating preprocessing, graph construction, model training, and evaluation stages for reproducibility

Adversarial Malware Detection and Evasion using the EMBER Dataset

Adapted an open-source EMBER detector implementation and integrated explainability-guided perturbation in malicious PE files

- Trained a static Windows PE malware classifier on the EMBER 2018 dataset (1.1M samples, ~2,300 features) using LightGBM
- Identified non-monotonic evasion behavior between 6 and 20 perturbations, where positive SHAP contributions increased (3.21 → 3.70) despite added noise, attributable to gradient-boosted trees activating higher-impact leaf combinations
- Implemented two functionality-preserving adversarial perturbation strategies (section injection and padding), achieving up to 37–38% reduction in absolute SHAP attribution across 50 perturbation iterations

GPT-2 Fine-Tuning with PyTorch & Hugging Face

Built an end-to-end pipeline to fine-tune a pretrained GPT-2 language model using PyTorch and Hugging Face Transformers in Google Colab

- Conducted hyperparameter tuning (learning rate, batch size, sequence length) to improve convergence and text coherence
- Modified training loop to manage attention masks, dataset preprocessing, and tokenization for contextualized fine-tuning
- Evaluated model performance using training loss metrics and qualitative analysis of generated text outputs

Experience

Semantic Shift Directed Research Occidental College

January 2026 - Present

Student Researcher Los Angeles, California

- Fine-tuned MacBERTh and BERT models on early modern texts using domain-adaptive masked language modeling (DAPT)
- Investigated semantic change detection by comparing contextual word embeddings across historical and modern English corpora
- Integrated Google Cloud Storage for automated checkpoint saving during pretraining, ensuring fault-tolerant training

Accenture

June 2025 - August 2025

Summer Analyst Los Angeles, California

- Built financial model projecting 25% revenue increase from digital payment adoption
- Collaborated with cross-functional teams across analytics and product to propose scalable solutions for customer outreach and analytics
- Translated customer data into clear, actionable insights to develop a client-facing slide deck

Cloud Computing Security and Privacy REU

May 2024 - July 2024

Student Researcher Boise, Idaho

- Researched adversarial attacks against intelligent detection models in collaboration with a team
- Synthesized literature on adversarial attacks to identify potential future research directions through a survey paper
- Presented findings to senior researchers and faculty during the 2024 Idaho Conference on Undergraduate Research (ICUR)

Occidental College

May 2023 - Present

Residential Advisor Los Angeles, California

- Mentored and supported a diverse community of 53 students
- Mediated conflicts and enforce campus policies to promote a healthy living environment
- Implemented career growth and drug awareness campaigns to enhance the residential experience